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Trade liberalization and domestic vertical integration: Evidence from China

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ABSTRACT

In this study we examine the effects of trade liberalization on domestic backward vertical integration in which a domestic upstream firm (target) is acquired by a domestic downstream firm. We first build a relationship-specific investment model to guide and provide insights to our empirical work. Then we take China's accession to the WTO as a quasi-natural experiment for trade liberalization to test the theoretical predictions. Consistent with the model, we find that a decrease in tariffs on the target industry's outputs reduces vertical integrations, but a decrease in tariffs on the target industry's inputs increases vertical integrations. The findings are robust to various specifications of the empirical model and measurements of the variables. We further show that underinvestment problem is an important mechanism to understanding the effects of tariff reductions on firms' organizational choices.

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1. Introduction

How do firms organize their production? Is it through market (outsourcing) or within firms (vertical integration)? This is a classic question in economics. This important question has also been asked in the field of international trade and foreign direct investment (FDI). How do firms organize their production *across countries*, through market or within firms (FDI)? On the one hand, the advancement of information and communication technology has changed the incentives for vertical integration. On the other hand, the decrease in transportation costs and reduction in trade barriers have altered firms' decisions on allocating their production across national boundaries. Theories of firm boundaries have been

introduced to address this issue. Most studies in this trade-FDI-organization literature focus on the effects of globalization on multinational corporations' (MNCs) decisions on offshoring and global vertical integration.¹ In contrast to that focus, the present paper explores how globalization affects outsourcing and vertical integration *within* a country's boundary, that is, by *domestic* firms, as opposed to MNCs. The empirical analysis of the paper helps address the concern of Acemoglu et al. (2010, p889) who conclude that “[d]espite a large theoretical literature on the determinants of vertical integration, the economics profession is far from a consensus on the empirical determinants of vertical integration in general.”

By definition, every vertical integration involves an upstream firm and a downstream firm. Every vertical integration executed via an acquisition has a target firm and an acquiring firm. In this study, we examine the effects of trade liberalization in the upstream industry on

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¹ See a brief literature review by Antras (2016).

domestic *backward* vertical integration, in which a domestic upstream firm is acquired by a domestic downstream firm.² We consider two types of tariff reductions. One is related to tariffs on the inputs that the target uses in its production, which we refer to as *input tariffs*. The other type relates to tariffs on the output of the target, which we refer to as *output tariffs*. We investigate how a cut in input tariffs and a cut in output tariffs affect the incentive of backward vertical integration, respectively.

We first propose a theoretical model to guide and provide insights to our empirical work. In the model, a domestic firm in the upstream industry makes relationship-specific investment that increases the value of the final product. Underinvestment is a typical outcome in this situation. Vertical integration helps solve the underinvestment problem but the benefit of vertical integration is affected by the outside option, which comes from the import of the upstream products. The analysis shows that when there is a reduction of output tariff in the upstream industry, the downstream firm benefits more from substituting its supplier with imported input, which makes vertical integration relatively less attractive. In contrast, an input tariff reduction in the upstream industry amplifies the benefits from vertical integration.

We take China's accession to the World Trade Organization (WTO) as a quasi-natural experiment for trade liberalization to examine its effects on vertical integrations. This approach has two advantages. First, China implemented drastic trade liberalization when it was accepted as a member of the WTO in 2001. The timing of China's accession to the WTO is commonly viewed as a shock. Second, reorganization of firms *within* China has become very active since the end of the last century. Fig. 1 plots the time trends of three types of mergers and acquisitions (M&As) related with China. During 1998–2014, a total of 18,220 M&As involving Chinese firms were completed. Among those, 67.9% or 12,371, are pure domestic M&As. The total value of all reported transactions is USD 741.2 billion and that of domestic M&As is USD 411.4 billion, accounting for 55.5% of the total value.³ As Fig. 1 shows, domestic M&As experience a sharp increase after 2001, while the increases of cross-border M&As are less significant.

We conduct our empirical analysis based on data from 1998 to 2007, with China's WTO entry around the middle of the period. We observe a large variation in tariff reductions across industries after the WTO entry, which allows us to employ the difference-in-differences (DID) method in our analysis. Our empirical results are consistent with the model prediction: output tariff liberalization *reduces* the number of (domestic backward) vertical integrations, while input tariff liberalization *increases* it. The findings are obtained based on our main regression model in which we control for a set of industry-level characteristics including the average age of firms, total size, degree of market competition, and time-varying international shocks. The findings are robust to various specifications of the empirical model and measurements of variables.

Although considerable empirical literature on the determination of vertical integration exists, empirical studies on the effects of tariffs on vertical integration are relatively new and still scant. As pointed out by Ornelas and Turner (2012, p31), “the relationship between tariffs and industrial structure has received very little empirical scrutiny”. The present paper contributes to this part of the literature, which includes

² When an upstream firm acquires a downstream firm, it is called forward vertical integration. The present study focuses on domestic backward vertical integration for two reasons. First, as pointed out by Acemoglu et al. (2010) and others, backward vertical integration is the most important alternative to non-integration. Backward vertical integration has also been a focus of other studies including Acemoglu et al. (2010) and Antras (2003). Second, in the case of China, backward vertical integration is indeed more prevailing than forward vertical integration. During 1998–2007, the total reported value of backward vertical integrations is USD 19.3 billion, accounting for 41.2% of the total value of all completed mergers and acquisitions, while forward vertical integrations account for 22.1%.

³ Not all M&As report their transaction value and thus, the reported value is only a fraction of the actual total value.

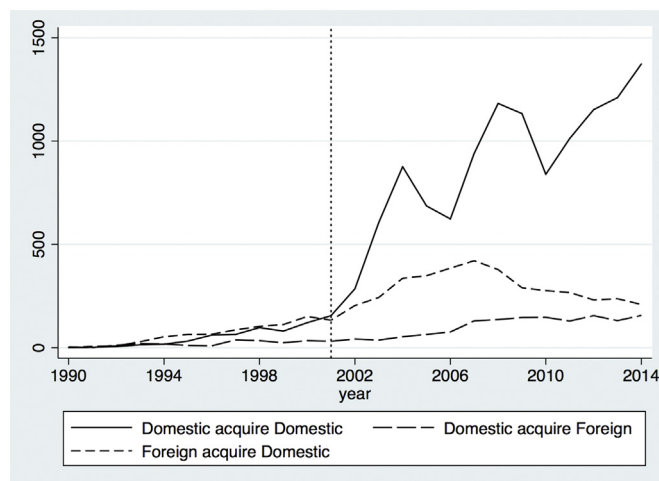


Fig. 1. Domestic and cross-border M&As related with China. Note: M&As are calculated based on the number of each type of M&As.

Alfaro et al. (2016) as the only existing study, to the best of our knowledge.⁴ Alfaro et al. (2016) build a simple model in which vertical integration increases productivity but is costly. Their model predicts that higher prices of final goods (the *downstream* firms' output) will make the upstream and downstream firms benefit more from vertical integration. They utilize tariff reductions as an exogenous shock to prices and use cross country data in their empirical analysis to test their theoretical prediction. Unlike (or as a complement to) Alfaro et al. (2016), we examine the effects on vertical integration by changes in the *upstream* industry's input and output tariffs. Moreover, our empirical findings can be explained using a model of relationship-specific investment and trade, which is built based on Ornelas and Turner (2008, 2012).⁵

Our paper is also related to existing studies on firm-level adjustments and performances in response to trade liberalization. Firm-level adjustments have many dimensions, including employment (Autor et al., 2013), total factor productivity (TFP) (Topalova and Khandelwal, 2011), quality upgrading (Amiti and Khandelwal, 2013), growth (Baldwin and Gu, 2009), innovation and R&D (Bloom et al., 2016; Liu and Qiu, 2016), product scope (Qiu and Zhou, 2013), and product mix (Mayer et al., 2014). These types of adjustments are *within-firm* adjustments. Our paper is about *inter-firm* adjustment (outsourcing or vertical integration), and hence, is also related to the literature reviewed by Antras (2016).

Some recent studies emphasize the importance of liberalization in intermediate input tariffs, as opposed to output tariffs. Those studies focus on the effects of intermediate input tariff reductions on firm productivity (Amiti and Konings, 2007), product scope (Goldberg et al., 2010), product quality (Fernandes and Paunov, 2013), and innovation (Liu and Qiu, 2016). Our paper adds a new dimension to this list: intermediate input tariff reductions affect domestic firms' organizational choice.

The paper is organized as follows. Section 2 provides a background of relevant laws, regulations and institutions in China. In Section 3, we present a conceptual framework. Section 4 introduces the econometric specification of the main model and describes the data. Section 5 reports the basic regression results and checks their validity and robustness. Section 6 concludes the paper.

⁴ In an empirical study, Breinlich (2008) shows that trade liberalization between Canada and the US has increased the domestic M&As in Canada. That paper, however, is not about vertical integration.

⁵ Extending Alfaro et al. (2016) model to tariff changes in the upstream industry can also generate predictions similar to our model (see online appendix). However, we have additional evidence (Section 5.3) to support our model.

2. Institutional background

2.1. China's accession to the WTO and trade liberalization

China began to apply for WTO entry in the late 1980s. After many rounds of negotiations, China finally joined in December 2001 and carried out a large cut in tariffs. Fig. 2 exhibits the tariff changes during the period of 1998–2007. We use the International Standard Industrial Classification (ISIC) to classify industries at four-digit level and calculate each industry's input and output tariffs. Output tariffs are directly available from the country's Effectively Applied Tariffs, called AHS tariffs, and input tariffs are calculated using output tariffs and the Chinese Input-Output Table (2002) (see details in Section 4.2). The left panel of Fig. 2 shows the means and standard deviations of the simple average output tariffs while the right panel shows those of the input tariffs. It is evident that during the pre-WTO period, 1998–2001, tariffs did not change much, staying at high levels, around 20% for output tariffs and 9% for input tariffs. In 2002, the first year of WTO entry, output tariffs dropped significantly from 18.67% to 14.10%, and so did input tariffs from 8.88% to 6.36%. Subsequently, both output and input tariffs continued to decrease and then remained stable after 2005 at around 10.42% for output tariffs and 4.86% for input tariffs. Significant variations in tariff levels and reductions across industries can be observed. The standard deviations of output tariffs are around 60% of the means while those of input tariffs are around 30% in the entire period.

Fig. 3 plots the tariff reductions of all industries during the post-WTO period, i.e., 2002 to 2007, against the corresponding industries' initial tariff levels in 2001. We can see strong positive correlations for both input and output tariffs: industries with high initial tariff levels experienced large tariff reductions in the post-WTO entry period.

In summary, the two figures present four features of tariff changes in China during 1998–2007. First, a sudden and drastic reduction in tariffs

took place in 2002. Second, substantial variations of tariff reductions occurred across industries. Third, across all industries, a strong and positive correlation between the initial tariff levels and the subsequent tariff reductions exists. Fourth, tariffs in the pre-WTO period were more or less constant. These features of tariff changes enable us to use the tariff levels of each industry in 2001 to proxy the degrees of tariff reductions. The validity of our identification strategy relies on the exogeneity of the key explanatory variable, i.e., the tariff levels in 2001. Bloom et al. (2016) and Liu and Qiu (2016) argue that the nature of China's accession to the WTO provides a quasi-natural experiment for such an identification. The sudden and drastic drop of tariffs in 2002 reflects the pure policy change because of China's accession to the WTO, and is unlikely to be driven by other confounding factors associated with vertical integration.

2.2. M&As and related regulations

M&As were not active in China until the late 1990s. According to Reuters' M&A database, the first domestic M&A in China was completed in 1987. In the early years, only a small number of M&As occurred, one in 1990, two in 1991 and five in 1992. Since then, M&A activities had continued to grow and became very active in the late 1990s. In 1998, the number of completed domestic M&As reached 100 in a year for the first time. The number increased dramatically after China's accession to the WTO, reaching its peak at 969 in 2007.

The first law on M&As promulgated in China is the *Interim Provisions on Enterprises Mergers and Acquisitions* (1989). The law provides guidelines and legal basis for conducting M&As in China. Subsequently, the government introduced a series of laws, provisions, regulations, guidance, and policies, some of which regulate M&A in general while some are specific to certain types of M&As.

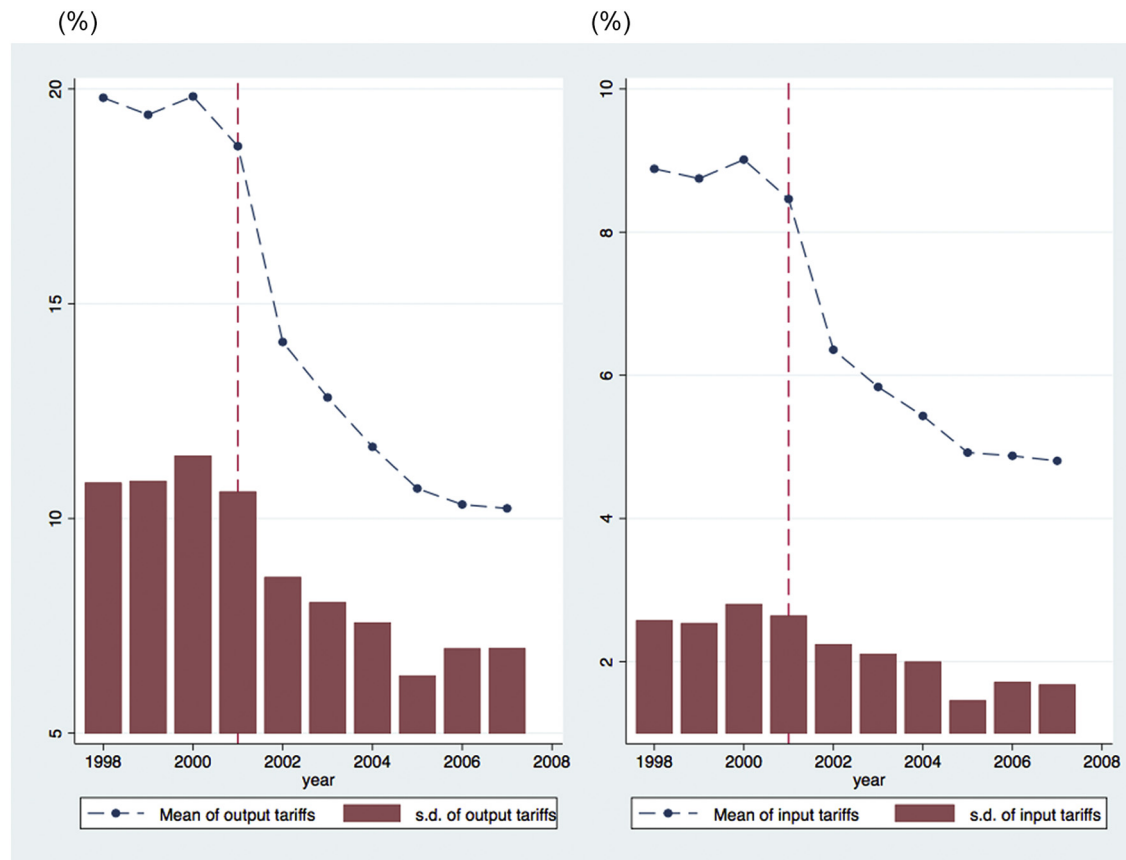


Fig. 2. China's output and input tariffs: 1998–2007.

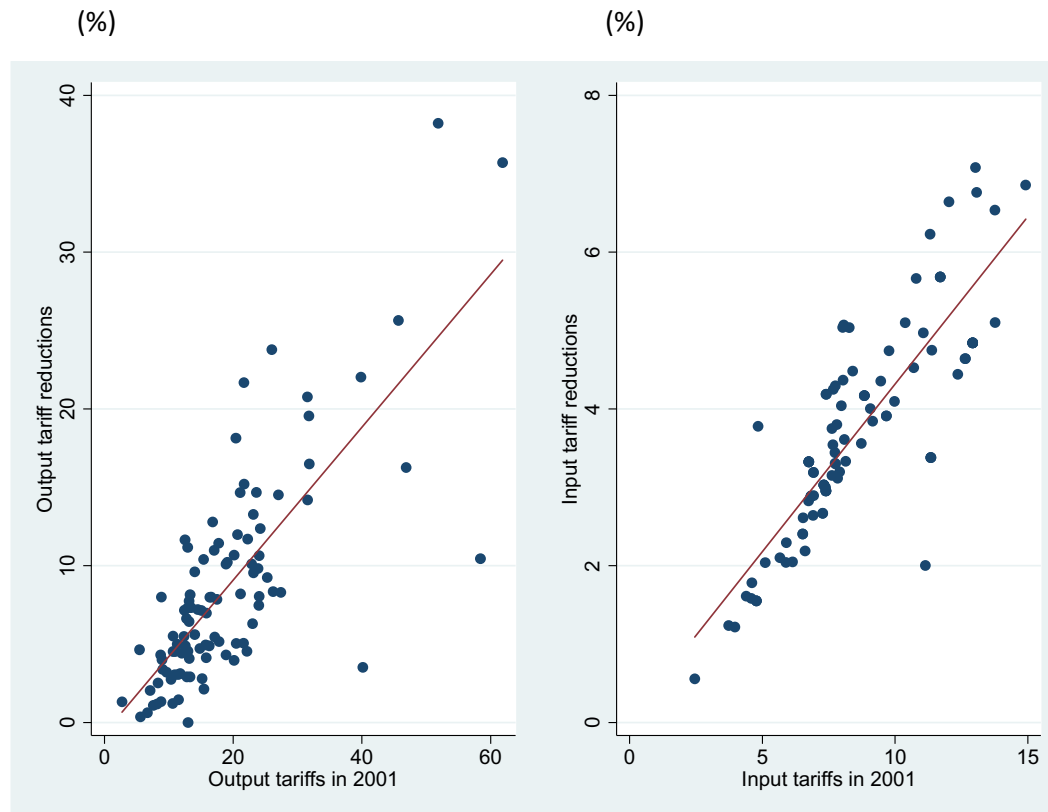


Fig. 3. Initial tariffs and tariff reductions across industries.

As for the general type of M&A laws and regulations, the *Anti-Monopoly Law of People's Republic of China* (2008) is the most comprehensive law on anti-monopoly in China. The first draft of the *Anti-Monopoly Law* was officially launched in 1994, but it only came into effect on August 1, 2008. Although M&As, including vertical integrations, are regulated directly by this law, our analysis is not affected because our data are from the sample period which ends in 2007, prior to the introduction of this law.

Other policies and regulations that are specific to certain types of M&As also exist. The first type is M&A policies on State Owned Enterprises (SOEs). China carried out SOE reforms mainly in the late 1990s and early 2000s. Reforms include privatization, M&As, and bankruptcy. Specific policies are *Notice on Several Issues in Trial of Merger and Acquisition and Bankruptcy of SOEs* (1996), *Notice on Relevant Issues Regarding the Selling of Small SOEs* (1999), and others related to foreign investors purchase of SOEs. These regulations may affect the incentives and outcomes of M&As involving SOEs. To insulate the effects of these regulations, we control for factors related to SOEs in our robustness checks.

The *Administration of the Takeover of Listed Companies Procedures* (2002) is specific to listed firms. Although the introduction of this regulation could affect takeover of listed companies differently from other types of takeovers, our identification strategy remains valid as long as the effects of this regulation across industries do not systematically coincide with the effects of trade liberalization. Nonetheless, we will check with a subsample of unlisted firms in our robustness regressions to control for possible effects of this regulation.

Another type of specific policies relates to foreign invested enterprises (FIEs) in China, such as *Provisions on Mergers and Divisions of FIEs* (1999), and *Interim Provisions on Mergers with and Acquisitions of Domestic Enterprises by Foreign Investors* (2003). These policies, however, do not have direct effects on the pure domestic M&As, which are the focus of the present study.

3. Model

We build a relationship-specific investment model based on Ornelas and Turner (2008, 2012) to understand how trade liberalization in the upstream industry affects backward vertical integration. To save space, we leave all the proofs to the online appendix.

Suppose that there are two domestic agents, called the (downstream) buyer and the (upstream) supplier. The supplier supplies a fully specialized widget to the buyer. The supplier can make an *ex-ante* relationship-specific investment $e \in [0, \bar{e}]$ to better adapt its widget to the buyer's need, but the investment is not contractible between the buyer and the supplier. The value of the specialized widget to the buyer is $V(e)$, which is increasing in the relationship-specific investment, with $V'(e) > 0$ and $V''(e) < 0$. The investment cost is $I(e)$, with $I'(e) > 0$. We normalize this cost function without much loss of generality: $I(e) = e$. To produce the widget, the supplier needs to import an input. Let τ be the tariff on the imported inputs, called the *input tariff*. We write the production cost of the widget as $C(\tau)$, with $C'(\tau) > 0$. We assume that the production cost is not extremely high: $C(\bar{\tau}) < V(0)$, where $\bar{\tau}$ is the upper bound of the input tariff τ .

The buyer can either use the specialized widget produced by the supplier, or import a standardized widget from abroad (offshoring).⁶ The foreign market of the standardized widget is competitive. If the buyer imports from the foreign market, it has to pay a price p and an import tariff t , where t is the tariff on widget, called *output tariff*. Assume that p is a random variable over $[0, \infty]$ following a distribution F . The randomness of p can be attributed to information asymmetry, or shocks in the foreign market. The standardized widget provides a deterministic value V_0 to the buyer. We assume that output tariff t is not prohibitive,

⁶ We assume single sourcing here. In the online appendix, we extend the model to consider dual sourcing and show that the main mechanism in this baseline model does not change.

that is, $t < V_0 - (V(\bar{i}) - C(\tau))$. This condition is derived from the case that when the price of the standardized widget p is zero, import is always the best choice for the buyer because $V_0 - t > V(\bar{i}) - C(\tau)$.

If the buyer buys from the supplier (*outsourcing*), they sign an arm's-length contract and bargain over the price through Nash bargaining. We assume that the bargaining power is α and $(1 - \alpha)$ for the supplier and buyer, respectively, and $\alpha \in (0, 1)$.

Instead of outsourcing to the supplier, the buyer can also vertically integrate the domestic supplier.⁷ There exists a fixed reorganization cost $K > 0$ in the case of vertical integration.

The timing of the game is as follows. At time 1, the buyer and supplier decide on integration or not. At time 2, if not integrated, the supplier chooses the investment level; if integrated, the integrated firm chooses the investment level. The foreign price of widget p is realized at time 3. At time 4, without integration, the independent buyer decides whether to buy the widget from the independent supplier or directly import the standardized widget from the foreign market; with integration, the integrated firm decides whether to produce the widget within the firm or directly import from the foreign market. At time 5, production and trade (if there is any) occur. Note that the buyer cannot commit *ex-ante* (i.e., before p is realized) to purchasing the domestic specialized widget or importing the foreign standardized one. Instead, it makes an efficient purchasing decision *ex-post*.

3.1. Non-integration

Suppose that the decision at time 1 is not to integrate (*non-integration*). Then, at time 4, if the buyer outsources to the supplier, they determine the transaction price of the widget $\tilde{p}$ through Nash bargaining. The payoffs are $u_b^1 = V(e) - \tilde{p}$ for the buyer and $u_s^1 = \tilde{p} - C(\tau)$ for the supplier. If the contract breaks down, the supplier receives nothing. As the investment cost is sunk, the supplier's payoff is $u_s^0 = 0$, which is its outside option. The buyer may import the standardized widget or merely stop producing its output, whichever yields a higher profit. Clearly, it will import the standardized widget if and only if $p + t < V_0$. Thus, the buyer's outside option is $u_b^0 = \max\{V_0 - p - t, 0\}$.

Price $\tilde{p}$ is the solution to the following Nash bargaining between the buyer and supplier: $\tilde{p}(t) = \arg \max_{\tilde{p}} (u_b^1 - u_b^0)^{1-\alpha} (u_s^1 - u_s^0)^\alpha$, whenever $u_b^1 > u_b^0$, $u_s^1 > u_s^0$. Hence, we have

$$\tilde{p}(t) = \begin{cases} \alpha[V(e) - (V_0 - p - t)] + (1 - \alpha)C(\tau) & \text{if } \phi(e, t, \tau) \leq p < V_0 - t, \\ \alpha V(e) + (1 - \alpha)C(\tau) & \text{if } p \geq V_0 - t, \end{cases}$$

where $\phi(e, t, \tau) \equiv (V_0 - t) - (V(e) - C(\tau)) > 0$.

At time 2, given the expected bargaining result, the supplier chooses its investment level e to maximize its expected profit. Assume that the second order condition (SOC) holds. Then, from the first order condition (FOC), we obtain the optimal investment level, e^n , under non-integration, which is defined by $\alpha V'(e^n)(1 - F(\phi(e^n))) = 1$. Thus, the supplier's expected profit is $u_s(e^n(t, \tau), t, \tau)$, the buyer's expected profit is $u_b(e^n(t, \tau), t, \tau)$, and the sum of the expected profits of the buyer and supplier is $U^n(t, \tau) = u_b(e^n(t, \tau), t, \tau) + u_s(e^n(t, \tau), t, \tau)$.

3.2. Vertical integration

Suppose that at time 1, the two firms decide to integrate. Then, at time 4, the integrated firm decides to produce the specialized widget by itself or import the standardized one from the foreign market. The operational profit of producing the widget by the integrated firm is $V(e) - C(\tau)$, and the one of importing is $V_0 - p - t$. The integrated firm chooses to produce by itself if and only if $V(e) - C(\tau) > V_0 - p - t$, which gives $p > \phi(e, t, \tau)$. Hence, the expected profit of making *ex ante* investment is given by $u_v(e, t, \tau) = \int_{\phi(e)}^{\phi(e)} (V_0 - p - t) dF(p) + \int_{\phi(e)}^{\infty} (V(e) - C(\tau)) dF(p) - e$.

At time 2, the integrated firm chooses its investment level e to maximize the above expected profit. Assume that the SOC for maximizing u_v holds. Then, from the FOC, we obtain the optimal investment level, e^v , as defined by $V'(e^v)(1 - F(\phi(e^v))) = 1$. The integrated firm's expected profit is $U^v(t, \tau) = u_v(e^v(t, \tau), t, \tau)$.

Lemma 1. The optimal investment level under vertical integration is larger than that under non-integration, i.e., $e^v > e^n$.

3.3. Organizational choice

Suppose that there is no wealth constraint and all agents are risk neutral. Then, by Coase Theorem, integration will happen if and only if the difference between the expected total profit under integration and that under non-integration is large enough to cover the reorganization cost K . We explore how tariff changes affect the comparison of the two organizational choices.

Define $\Delta U(t, \tau) = U^v(t, \tau) - U^n(t, \tau)$, which captures the incentive for integration. We can show that

$$\frac{\partial \Delta U(t, \tau)}{\partial t} = [F(\phi(e^n)) - F(\phi(e^v))] - \frac{1 - \alpha}{\alpha} \frac{\partial e^n}{\partial t} \quad (1)$$

and

$$\frac{\partial \Delta U(t, \tau)}{\partial \tau} = -C'(\tau)[F(\phi(e^n)) - F(\phi(e^v))] - \frac{1 - \alpha}{\alpha} \frac{\partial e^n}{\partial \tau}. \quad (2)$$

Proposition 1. Suppose that the value function $V(e)$ is sufficiently concave (i.e., $V''(e) < 0$ and $|V''|$ is large). Then, a decrease in the upstream industry's output tariff will lead to fewer vertical integrations ($\frac{\partial \Delta U(t, \tau)}{\partial t} > 0$), while a decrease in the upstream industry's input tariff will result in more vertical integrations ($\frac{\partial \Delta U(t, \tau)}{\partial \tau} < 0$).

The intuition is as follows. We focus on how tariffs affect the comparison between integration and non-integration. We first look at output tariff. There are two forces at work. On the one hand, at any given tariff level, the buyer's preference of using imported widget over the domestic one is stronger under non-integration than under integration because of the underinvestment associated with non-integration (Lemma 1). That is, the buyer relies more on imported widget under non-integration than under integration, and an output tariff cut benefits the buyer more under non-integration than under integration. Thus, an output tariff cut increases the attractiveness of non-integration, which is captured by the first term in Eq. (1) (i.e., $[F(\phi(e^n)) - F(\phi(e^v))] > 0$). On the other hand, when the output tariff decreases, the non-integrated domestic supplier anticipates that the buyer is less likely to purchase its product, and thus would decrease its relationship-specific investment, which is captured by the second term in Eq. (1) (i.e., $\frac{1 - \alpha}{\alpha} \frac{\partial e^n}{\partial t} > 0$). However, if $V(e)$ is concave enough, the impact of tariff t on investment

⁷ In a different setting in which foreign suppliers can also make relationship-specific investment (Ornelas and Turner, 2008), we should also allow the possibility of the buyer vertically integrating a foreign supplier. We rule out this possibility to have a sharper focus. First, Ornelas and Turner (2008) have already studied the case of foreign integrations. Second, our focus on domestic vertical integration is more in line with the reality of China. The occurrence of Chinese firms acquiring foreign firms are relatively rare during our sample period (1998–2007). In this period, the total number of vertical integrations in which Chinese firms acquire foreign firms is only 121, while the total number of domestic vertical integrations is 1540.

e^n is not strong, and the first effect will dominate the second effect, which results in $\frac{\partial \Delta U(t, \tau)}{\partial t} > 0$.

As for input tariff, on the one hand, a decrease in the tariff will reduce the domestic supplier's cost of producing the widget. **This makes the buyer more likely to use the domestically produced widget, which helps increase the supplier's incentive to make relationship-specific investment.** Thus, integration becomes more attractive, which is captured by the first term in Eq. (2) (i.e., $-C'[F(\phi(e^n)) - F(\phi(e^v))] < 0$). On the other hand, the decrease of input tariff also induces more relationship-specific investment under non-integration because the likelihood of using domestically produced intermediate goods is increased as shown by the second term in Eq. (2) ($\frac{1-\alpha}{\alpha} \frac{\partial e^n}{\partial \tau} < 0$). When $V(e)$ is concave enough, **the second effect is small, and so a decrease in the upstream industry's input tariff leads to more vertical integrations.**

To put our model in perspective of the related literature, we compare it to some existing models. Ornelas and Turner (2008) analyze exactly the opposite issue: the buyer's decision to integrate or offshore to a foreign supplier, with passive domestic suppliers as the outside option. Ornelas and Turner (2008) find that lower tariff (i.e., output tariff t as in our model) promotes (cross-border) vertical integration. The trade liberalization in input tariff in our model (i.e., τ) is absent in their model. Ornelas and Turner (2012) is a theoretical analysis on optimal trade policy in the presence of endogenous organizational choice of firms. Input tariff faced by the domestic supplier is not considered in their model either. Alfaro et al. (2016) investigate how the price of the final product (the downstream industry's output tariff in our model) affects vertical integration. In contrast, our model complements theirs in the sense that we explore how the prices of upstream output and input affect vertical integration.⁸

In the rest of the paper, we test the prediction of Proposition 1 using Chinese data: output tariff reductions would discourage backward vertical integrations, whereas input tariff reductions would promote vertical integrations. Two main issues will emerge in the test. The first is about the sufficient condition (sufficiently concave value function V) and the second is about the causal effects of tariffs on vertical integration. Although concavity is a common assumption in this literature, we are not able to find a good way and data to test it. Thus, our effort is to focus on the causality issue, which we address by utilizing the fact that China's WTO entry represents exogenous changes of input and output tariffs.

4. Econometric specification and data

4.1. Empirical model

To identify the causal impact of trade liberalization on vertical integration, we exploit the quasi-natural experiment based on China's accession to the WTO in 2001 to conduct a DID analysis. Our data covers the period 1998–2007, which includes both the pre-WTO (1998–2001) and post-WTO periods (2002–2007). China's WTO accession resulted in a large scale tariff reductions across all industries, but with a significant degree of variations. This feature enables us to conduct a DID estimation that examines the difference between the changes in vertical integration activities in industries with larger tariff reductions (the treatment group) and the corresponding changes in industries with smaller tariff reductions (the control group).

⁸ Alfaro et al. (2016) also mention the effect of the price of intermediate inputs on the organizational choice (see footnote 14 of their paper). They argue that the effect depends on whether the firm is a net buyer or seller of the intermediate inputs. Our paper takes a different approach in examining the organizational form, which is determined by the two opposing effects associated with the hold-up problem.

Accordingly, we construct our basic econometric specification as

$$V_{it} = \beta_1 \cdot WTO_t \times OutT01_i + \beta_2 \cdot WTO_t \times InT01_i + \mathbf{X}_{it}'\Gamma + \lambda_t + \lambda_i + \varepsilon_{it} \quad (3)$$

In Eq. (3), V_{it} is the total number of acquisitions with the targets in industry i (at ISIC four-digit level) and the acquirers from the downstream of industry i , in year t . This dependent variable is the count of vertical integrations, specified for the upstream industry. WTO_t is a dummy variable indicating the years after 2001, the post-reform period, i.e., it takes 0 for all years before 2002, and 1 for all years from 2002 onwards. $OutT01_i$ is the level of output tariff of industry i in year 2001. Similarly, $InT01_i$ is the level of input tariff of industry i in 2001. Because the tariff levels in 2001 are positively correlated with the degrees of tariff reductions associated with the WTO accession, $OutT01_i$ and $InT01_i$ capture the extents of output and input tariff reductions (treatment intensity), respectively. We use year fixed effect λ_t to control for time trends and all other possible time-specific shocks. We include industry fixed effect λ_i to control for time-invariant industry characteristics which might be associated with industry-level vertical integrations. We also include a set of time-varying industry characteristics that might affect vertical integration. This set of variables, represented by $\mathbf{X}_{it}'$, includes the average age of the firms in the industry, the size of the industry, the degree of market competition of the industry, and the time-varying international shocks to the industry. The last term in Eq. (3), ε_{it} , is the robust standard error term clustered at the ISIC four-digit industry level. Because the dependent variable is a count variable, we use the fixed-effects Poisson model to obtain the estimation results.

The reliability of our estimation depends on the validity of our model specification, that is, whether our regressors of interest are independent of the error term conditional on all control variables. China's WTO accession is a long negotiation process and the timing of accession is highly political and unpredictable, which implies that our WTO_t variable is very likely to be exogenous (Bloom et al., 2016). However, the degree and procedure of tariff reductions across industries may be an outcome of negotiation and therefore, may not be completely exogenous. To deal with this problem, instead of using the actual tariff reductions, we purposely use the initial levels of tariff ($OutT01_i$ and $InT01_i$) before WTO accession to measure the treatment intensity.⁹ This measure helps alleviate the potential concern of endogeneity because the initial tariff levels are basically fixed during the pre-WTO period and thus are largely pre-determined before our sample period. In addition, we follow Goldberg et al. (2010) and Liu and Qiu (2016) to check whether the levels of output and input tariffs in 2001 and their actual changes after the WTO entry are related to industry performances in the pre-WTO period. We find that vertical integration activities and other industrial performances in the pre-WTO period are not correlated with the initial levels of output and input tariffs in 2001 (see online appendix Table A1), which lends a further support to our model specification.

If our theory is correct, then β_1 is negative and β_2 is positive. Because our theory is about the qualitative (rather than quantitative) aspect of the effect of tariff changes on vertical integrations, our DID approach is appropriate to do the test and β_1 and β_2 will capture the effects.

4.2. Construction of the key variables

Our empirical analysis has two key variables: vertical integrations and tariffs. In the main analysis, we measure vertical integrations using the number of acquisitions by firms from downstream industries with the targeting firms from a specific upstream industry i (V_{it}). We

⁹ We also run regressions with tariff levels, i.e. $V_{it} = \beta_0 + \beta_1 \cdot OutT_{it} + \beta_2 \cdot InT_{it} + \mathbf{X}_{it}'\Gamma + \lambda_t + \lambda_i + \varepsilon_{it}$. The signs of the estimates are consistent with our DID estimates, but less significant. One possible reason is the endogeneity problem associated with the evolution of tariffs. That is also why our study and others prefer using initial tariffs in the DID specification.

will check the robustness of our results using the value of those acquisitions for VI_{it} .¹⁰ In either case, we need to first define and identify upstream and downstream. For any given industry, defining the upstream and downstream industries is straightforward theoretically, but is less so in practice. We follow [Antras et al. \(2012\)](#) to identify upstream and downstream based on the upstreamness index (U_i). A higher U_i means that industry i is at a more upstream stage in the value chain.

We use China's Input-Output Table (2002) to calculate the upstreamness index and match it with ISIC (Revision 3) industries.¹¹ For each M&A, if the target is at a more upstream position (higher upstreamness index) than the acquirer, this M&A is defined as a backward vertical integration. We also use other measures of vertical integration in the robustness checks.

Trade liberalization in each industry includes input and output tariff reductions. Our output tariffs, $OutT_{it}$, are AHS simple average tariffs at ISIC four-digit industry level, which are available directly from the World Integrated Trade Solution (WITS). We construct input tariffs following [Liu and Qiu \(2016\)](#). Specifically, input tariff for industry i (according to the classification in Input-Output Table) in year t is defined as the weighted average of tariffs on goods used as inputs for industry i , that is, $InT_{it} = \sum_j CostShare_{ij} \cdot OutT_{jt}$, where InT_{it} is the input tariff of industry i in year t , $OutT_{jt}$ is the output tariff of industry j in year t , and $CostShare_{ij}$ is the cost share of industry j in producing one unit of output of industry i . $CostShare_{ij}$ is based on China's Input-Output Table (2002). Thus, the initial input tariff of 2001 in Eq. (3) is obtained as $InT01_i = InT_{i2001}$.¹²

4.3. Data

Our empirical analysis relies on three important data sources. The first data source is the M&As database from Thomson Reuters SDC Platinum (SDC). This database includes transaction-level data of M&As involving at least 5% ownership of the target and a transaction value of one million U.S. dollars or more, or where the value of the transaction is undisclosed, all around the world. The database provides information on the targets and acquirers, such as firm name, country, industry, parent firm information, primary business, and many financial characteristics. It also provides information on the M&A deal, such as the announced time, status, completion time, value of transaction, and share of transaction.

We calculate the value of our vertical integration variable (VI_{it}) based on the SDC database. We define a firm's industry according to its primary business.¹³ Because our VI_{it} is the number of domestic backward vertical integrations, we extract all transactions from SDC database with both acquirer and target nations stated as "China" in each

transaction (excluding Hong Kong, Macao, and Taiwan for the purpose of pure "domestic firms" by convention), with targets in manufacturing sector and acquirers from targets' downstream industries. We calculate the total number of backward vertical integrations targeting industry i at year t as our main measure of VI_{it} .

The second data source is the WITS, which provides import tariff data of many countries including China. As pointed out earlier, we use the AHS simple average tariff data at ISIC four-digit product level, which is available directly from the WITS.¹⁴

The third data source is the Above-Scale Industrial Firms Panel (ASIFP), which is available for the time period 1998–2007. The ASIFP is provided by China's National Bureau of Statistics (NBS). It covers all SOEs and non-SOEs with annual sales of at least 5 million RMB in China's mining, manufacturing, and utilities industries. The number of firms in this database varies from over 140,000 in the late 1990s to over 336,000 in 2007. Firms are from all 31 provinces and directly-administrated municipalities in China. We confine to firms from manufacturing industries.¹⁵ The ASIFP contains detailed information of each firm, including the firm's official name, industry, location, and ownership. It also contains many operation and performance items of each firm such as age, employment, capital, intermediate inputs, and new product sales. We clean the ASIFP data by dropping observations according to the basic rules of Generally Accepted Accounting Principles. We drop firms with fewer than eight workers because those firms are subject to a different legal regime ([Brandt et al., 2012](#)), and firms with obviously wrong establishment years such as 1900 and before. With the above effort, we obtain a dataset containing a set of time-varying industry-level control variables.

We merge the above three databases to construct our dataset at the ISIC four-digit industry level.¹⁶ The starting time of our dataset is 1998, which is the earliest available year from the ASIFP data. The ending time is 2007, which is subject to data availability and also chosen to avoid potential biases caused by the worldwide financial crisis in 2008. In our merged dataset, we have data for 102 ISIC four-digit industries in 1998–2007. We present the descriptive statistics of vertical integrations and tariffs across ISIC two-digit industries in online appendix Table A2, which shows substantial variations across industries in the changes of vertical integrations and tariffs.

5. Econometric analysis

5.1. Basic results

[Table 1](#) reports our basic regression results. In all regressions, the dependent variable is the number of backward vertical integrations targeting at industry i . We find consistent results that output tariff reduction in an industry significantly *decreases* the industry's vertical integrations, and input tariff reduction significantly *increases* the number of vertical integrations.

¹⁰ As argued by [Breinlich \(2008\)](#), using the count instead of total value of transactions has at least two advantages. First, the count measure is more readily available in the database because some transactions do not publish transaction values. Second, the distribution of transaction values is quite dispersed.

¹¹ We use the IO table of 2002 because China's IO table is available every five years and 2002 is in the middle of our sample and close to the time of China's WTO entry.

¹² Since the input tariff is constructed from the output tariff, multicollinearity problem is a potential concern in our regression model. However, as stated by [Wooldridge \(2016\)](#), multicollinearity problem is not really well-defined. It mainly leads to a larger estimated standard error, causing the estimated coefficients insignificant and unstable. Our robustness checks in later sections suggest that multicollinearity is not a severe issue in our settings because our results are highly significant and stable. We also calculate the variance inflation factor (VIF) to test the multicollinearity based on our baseline specification. The VIFs of our two key variables are 3.99 and 3.85 respectively, smaller than the commonly used cutoff value 10, which also suggests no severe multicollinearity.

¹³ SDC also provides data on other business lines of a firm. For multi-sector firms, we can calculate the average upstreamness of each firm to define vertical relationship between any pair of firms. The results are not different much from those we obtain based on a firm's primary business. Hence, we choose to follow the literature to define a firm's industry according to its primary business. Nevertheless, we are aware of the problem that we cannot accurately tackle the issue associated with multi-sector firms as we do not know which business line of the target is acquired.

¹⁴ We use tariff data instead of Non-Tariff Barriers (NTB) because the comparability of NTB is problematic and NTB data are more difficult to acquire. In contrast, tariff data are price-based measures, more comparable across sectors and time. Moreover, compared to other indirect measures such as import penetration rates, tariff cuts are a direct policy instrument and suffer less from endogenous problems ([Breinlich, 2008](#)).

¹⁵ We include all manufacturing industries except the tobacco industry, which is controlled by a monopoly SOE, the China National Tobacco Corporation (CNTC). CNTC runs the whole supply chains of the tobacco industry, covering plantation, allocation, manufacturing, distribution, and importing and exporting. Consequently, there is no issue of change in vertical structure in this industry. However, including tobacco industry in the analysis does not affect our results.

¹⁶ All M&A transactions in SDC are classified into SIC four-digit industries based on the primary field of the target company or acquired division. We convert SIC four-digit industry to ISIC four-digit industry using the correspondence table provided by UNSTAT. Variables from the ASIFP database are classified by Chinese Industry Classification, which is modified from the ISIC (Revision 3). Thus, we also need to reclassify firms into ISIC four-digit industries.

Table 1
Basic results.

	(1)	(2)	(3)	(4)	(5)
WTO · OutT01	−2.0530*** (0.7768)	−2.0083*** (0.7441)	−1.9408** (0.7593)	−1.9214** (0.7531)	−1.9005** (0.7447)
WTO · InT01	2.7312*** (0.9370)	2.5687*** (0.9559)	2.5675*** (0.9564)	2.5324*** (0.9497)	2.5085*** (0.9326)
logAge		−1.4531*** (0.5575)	−1.2212 (0.7628)	−1.2388 (0.7701)	−1.2447 (0.7657)
logN			0.2133 (0.4487)	0.2418 (0.4722)	0.2132 (0.4586)
HHI				1.8905 (5.0086)	1.0824 (4.8627)
India&Brazil					0.0319** (0.0161)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	580	580	580	580	580

Note: Robust standard errors clustered at ISIC four-digit industry.
 ** p < .05.
 *** p < .01.

Specifically, column (1) presents the basic results with only industry and year fixed effects being controlled. We find a significant and negative estimate for $WTO_t \times OutT01_i$, and a significant and positive estimate for $WTO_t \times InT01_i$. These findings indicate that after China's accession to the WTO, firms in industries that experienced larger output tariff reductions are less likely to be integrated by their downstream firms, but firms in industries that experienced larger input tariff reductions are more likely to be integrated by their downstream firms. To check whether these findings are obtained due to some omitted driving forces, we further control for time-varying industry characteristics that might be related to the decision of vertical integration.¹⁷

First, the life-cycle status of each industry may matter for vertical integrations, and thus in column (2) we add the logarithm of average firm age of industry i to the basic regression. We find that the sign and significance of the key estimates (i.e., output and input tariffs) are not affected by firm age. The estimate of firm age is negative and significant. When additional control variables are included in columns (3)–(5), the sign of firm age estimate remains negative but no longer statistically significant. Younger firms may be more dynamic and grow faster, and thus attract more downstream firms to vertically integrate them.

Second, the availability of potential targets in each industry is likely to be related to the number of M&As. In a larger industry, more firms are available as potential targets for vertical integration. This fact is especially true because our dependent variable is the count of vertical integration, not adjusted by industry size. The number of firms in an industry may also reflect the competition of the industry, which in turn may affect the incentives for vertical integration. To control for these effects, we add a new variable, $logN_{it}$ which is the logarithm of total number of firms in each industry in each year, to the basic regression in column (3). We find that the estimate of industry size is positive, as expected, but not statistically significant. More importantly, our main results (effects of tariff reductions) remain unchanged.

Third, although the number of firms in an industry is related to competition, the Herfindahl-Hirschman Index (HHI) is a better measure of competition. Fiercer competition may induce firms' restructuring activities. Accordingly, we add HHI to our regression to control for competition in each industry. This control also helps address the concern that the effect of China's trade liberalization on vertical integration is due to the change in the level of competition. The results are reported in column (4). The effect of HHI is positive but statistically insignificant, and

the coefficients of our regressors of interest are not affected. Thus, competition is not an important factor for explaining China's vertical integration.

Finally, although we include industry and time fixed effects in our specification to control for time-invariant industry characteristics and general time trends of vertical integration, vertical integration might also be affected by omitted time-varying international industry shocks (e.g., global technology shocks). Hence, to control for those shocks, we follow Breinlich (2008) to include the total number of completed vertical integrations in each industry in some other countries. In particular, we choose India and Brazil because they are similar to China in many dimensions as a large developing economy.¹⁸ The premise is that if international factors that affect the world vertical integration activities in large developing countries exist, they are likely to be similar for China, India, and Brazil. Furthermore, no significant trade liberalization had been carried out in India and Brazil around the time of China's entry to the WTO, which makes the number of vertical integrations in these two countries a better proxy for international industry shocks. Hence, we add to our regression $India \& Brazil_{it}$, the total number of backward vertical integrations in India and Brazil in industry i and year t . The results are in column (5). The estimate of $India \& Brazil_{it}$ is positive and statistically significant, which shows indeed the existence of international industry shocks. However, our key estimates (tariffs) remain unchanged.

Our back-of-envelope calculation shows that the economic magnitude of the impact of trade liberalization on backward vertical integrations is meaningful. Taking the baseline results in column (5) of Table 1 as an example, one log point change of output and input tariffs leads to −2.0530 and 2.7312 log points change of vertical integrations, respectively. Hence, moving an industry from the 25th percentile of initial level of output tariff to the 75th percentile decreases the vertical integration incident rate by 62.03%, while moving an industry from the 25th percentile of initial level of input tariff to its 75th percentile increases the incident rate of backward vertical integration by 97.61%. For a specific industry, the net impact of trade liberalization is then the sum of these two opposite effects as trade liberalization generally reduces both input and output tariffs.

5.2. Validity and robustness checks

The reliability of our estimates obtained earlier depends on the validity of our DID specification. In this subsection, we conduct a series of tests on the validity of the DID and perform some tests on the robustness of the results.

5.2.1. Validity of DID

5.2.1.1. Comparability of groups. In Section 5.1, we estimate the average treatment effect of the output tariff cut and input tariff cut. The treatment effect is the difference between the treatment and control groups in their average differences in backward vertical integrations between the pre-WTO and post-WTO periods. We now check the differences with a more flexible model specification. This check helps answer two questions: whether the two groups have comparable trends before the WTO accession; and whether the divergence of trends emerges after the WTO accession. To address these questions, we conduct the flexible estimation by replacing the WTO dummy with a vector of year dummies λ_t , indicating the years from 1998 to 2007 with year 2001 as the omitted reference year. This specification imposes little timing structure on the model. Thus, we are simply examining how the

¹⁷ The fixed effect Poisson estimation automatically drops industries with no vertical integration during the entire sample period, so we have 580 observations in these regressions. In the robustness check, we show that including those dropped industries without vertical integrations does not change our results.

¹⁸ We have conducted (but not reported in this paper) comparisons of the three countries in many dimensions, including population, economic growth, economic structure, trade, and number and distribution of backward vertical integrations. They all have similar patterns. China is more similar to India than Brazil. Using India alone to proxy the global trend gives qualitatively the same results.

difference in vertical integrations between industries with different tariff reduction levels varies over time.

We plot the flexible estimation results, i.e., the estimated coefficients of the interaction terms $\lambda_t \times OutT01_i$ and $\lambda_t \times InT01_i$, and also the means of the coefficients in Fig. 4. Two clear patterns emerge. First, in the pre-WTO period the coefficients are very unstable and the means of the coefficients are very close to zero. Second, in the post-WTO period the coefficients of output (input) tariff are consistently and significantly negative (positive), showing significant differences from those of the pre-WTO period. These results support our DID specification.

5.2.1.2. Expectation of WTO entry. Although the process and timing of China's WTO accession are highly political and widely regarded as unexpected, we still want to check whether firms somehow could anticipate the WTO accession time. If they had anticipated it, they would have prepared and responded rationally in 2001 just as they did after 2001. To address this issue, we add the interaction terms $\lambda_{2001} \times OutT01_i$ and $\lambda_{2001} \times InT01_i$ to Eq. (3). The regression results are reported in column (1) of Table 2. It shows that both the coefficients of $\lambda_{2001} \times OutT01_i$ and $\lambda_{2001} \times InT01_i$ are insignificant, which implies the absence of the expectation effect and the treatment and control groups do not show differential expectation. Moreover, the coefficients of the key regressors, $WTO_t \times OutT01_i$ and $WTO_t \times InT01_i$, remain significantly negative and positive, respectively.

5.2.1.3. Industry-specific trends. In our DID specification, we implicitly assume that conditional on $(X_{it}', \lambda_t, \lambda_i)$, our treatment and control groups each have a common trend in the pre- and post-WTO periods. This assumption enables us to use vertical integrations of the control group as the counterfactual of the treatment group in the post-WTO accession period. Differencing out the trend of the control group allows us to isolate the effect of tariff reductions on vertical integration. However, if industries are affected differently by confounding industry-time specific factors, they might have different trends in the pre- and post-WTO periods. One way to solve this problem is to allow industries to have different time trends in the pre- and post-WTO periods by introducing a vector of industry-WTO dummies, $\sum_i \lambda_i \times WTO_t$. However, this method is not applicable to our model because our key variables are defined at exactly the same ISIC four-digit industry level. Alternatively, we control for industry trends in a more aggregate level, ISIC two-digit industry level, by adding a set of dummies, $\sum_i ISIC2_i \times WTO_t$, to our regression model.

Results in column (2) of Table 2 show that after controlling for changes of industry-specific trends, the effects of input and output tariffs are qualitatively the same and quantitatively larger.

5.2.1.4. Other simultaneous reforms. Simultaneous reforms might contaminate the validity of our DID estimation. If there are other reforms introduced around the time of China's WTO accession and affecting our treatment and control groups differently, then the effects of those reforms would also be picked up in our DID estimates. Indeed, several reforms took place in the early 2000s and thus we want to control for any possible confounding effects.

First, the SOE and FDI reforms. These reforms started before China's WTO accession, but accelerated afterwards. The new SOE reform resulted in large-scale privatization, close-down of small SOEs, and efficiency improvement of the surviving (large) SOEs. The new FDI regulations relaxed the entry requirement for foreign investors and reduced the range of industries restricted to foreign investment. These two reforms may affect the market structure and competition which may affect the incentives for domestic vertical integration. We do not have perfect measures to represent these reforms. However, One important consequence of these reforms is the change of the number of SOEs and FIEs in each industry. Therefore, we include two additional variables in our DID estimation to control for potential influences from the two policy reforms: $SOEShare_{it}$ (measured by the ratio of the number of SOEs over the total number of domestic firms) and $FIEShare_{it}$ (measured by the ratio of the number of FIEs over the total number of firms). The regression results are reported in column (3) of Table 2, which shows that including policy reforms on SOE and FIE does not affect our estimates of the output and input tariffs. Hence, we can rule out the possibility that our results are caused by simultaneous SOE and FIE reforms.

Second, market entry regulation. Although market entry is regulated in certain industries in China, especially in the early period of our sample, the government has already relaxed the entry regulations to certain degree. The Report of the Sixteenth Party Congress in 2002 clearly states that "Monopoly industries should carry out reforms to introduce competition mechanisms... We should expand the areas for the market access of domestic non-governmental capital and adopt measures with regard to investment, financing, taxation, land use, foreign trade and other aspects to carry out fair competition." This reform may affect industrial reorganizations, especially in those entry-regulated industries. We calculate the logarithm of the total number of non-SOE entries in each industry, $NonSOE_Entry_{it}$, to measure the intensity of entry regulations across industries. The results in column (4) show that the coefficient of $NonSOE_Entry_{it}$ is negative but not statistically significant. The effects of tariffs on vertical integration are not affected by inclusion of this variable.

Third, M&A administration. The government promulgated the *Administration of the Takeover of Listed Companies Procedures* in 2002, which is directly related to M&As. This is a general regulation on

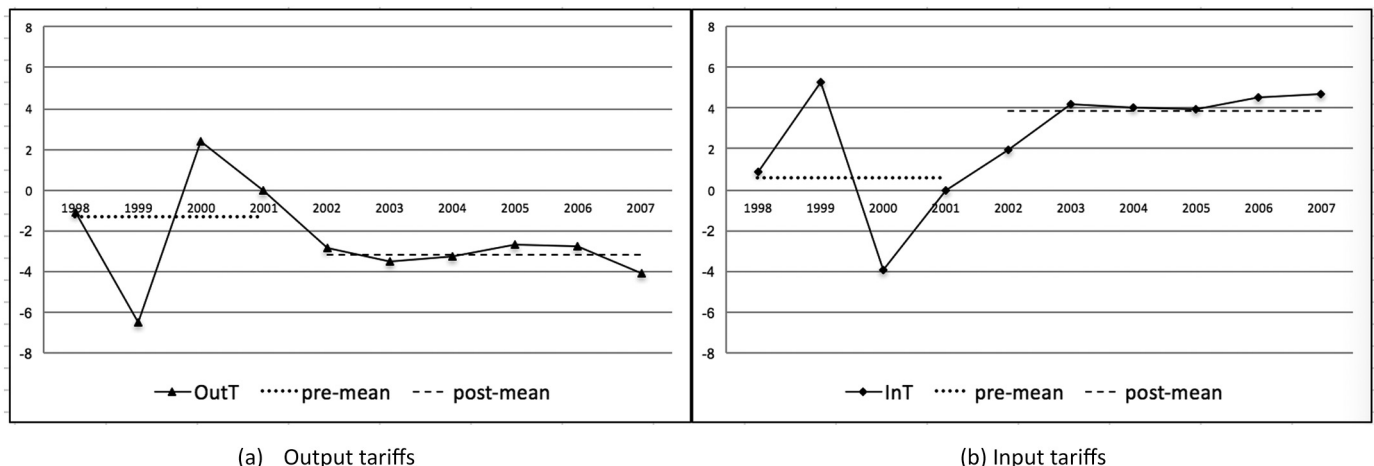


Fig. 4. Flexible estimation.

Table 2
Validity.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Expectation	Industry trend	SOE&FDI reforms	Entry regulation	Unlisted	All simultaneous reforms	Two-period
WTO·OutT01	−1.6433** (0.7403)	−2.8787*** (0.8203)	−1.7364** (0.7261)	−1.9008** (0.7488)	−2.2895*** (0.8681)	−2.0335** (0.8476)	−1.8705** (0.7496)
WTO·InT01	2.4161** (1.0691)	4.1566** (1.7494)	2.3637** (0.9336)	2.4735*** (0.9564)	2.5585** (1.1893)	2.2657* (1.1916)	2.6291*** (1.0184)
y2001·OutT01	0.6117 (1.2543)						
y2001·InT01	−0.1864 (1.9967)						
SOEShare			6.2478** (3.0718)			9.7189*** (3.5521)	
FIEShare			−2.4014 (3.3995)			−3.8652 (4.2324)	
NonSOE_Entry				−0.1540 (0.3029)		−0.3693 (0.4966)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	580	580	580	580	490	490	116

Note: Robust standard errors clustered at ISIC four-digit industry. Controls include logAge, logN, HHI, and India&Brazil.

* p < .10.

** p < .05.

*** p < .01.

takeovers of *listed* companies across all industries and thus is unlikely to confound with the differential effects of trade liberalization on vertical integration of *all* firms. However, to exclude the possibility of its confounding effect, we run the regression with the subsample of *unlisted* firms. Specifically, we exclude all vertical integrations with listed companies as targets in our data and re-count the total number of backward vertical integrations as our new dependent variable. The results are presented in column (5), which indicates that, after excluding all takeovers of listed companies, the effects of output and input tariffs on vertical integration remain unchanged.

In column (6) of Table 2, we include all the above-discussed simultaneous reforms in the regression. The consistently significant coefficients on output and input tariffs indicate that our specification should be valid in capturing the effects of trade liberalization.

5.2.1.5. Two-period sample. Serial correlation of error terms might be a concern in DID estimations. To alleviate this problem, in all regressions we cluster the error term at the industry level, which allows serial correlation and heteroscedasticity across years. Now, we follow Bertrand et al. (2004) to collapse the data into two periods, pre- and post-treatment periods, which helps reduce serial correlation. This method also helps smooth the data (as there are many zeros). Specifically, we use the total number of vertical integrations of each industry in the pre- and post-WTO periods as our dependent variable, and use the mean of all explanatory and control variables in pre- and post-WTO periods as our independent variables. Based on this new two-period panel data, we rerun our baseline specification and report the results in column (7) of Table 2. The estimates of interest remain robust.

Table 3
Robustness.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Subsidy	Exports	Foreign tariff	RCA	TFP	Upstream liquidity	All
WTO·OutT01	−1.8991** (0.7633)	−1.8881** (0.7452)	−1.9224** (0.7518)	−1.7280** (0.7667)	−1.8082** (0.7605)	−1.9099*** (0.7236)	−1.7403** (0.7745)
WTO·InT01	2.4644*** (0.9442)	2.4793*** (0.9448)	2.5270*** (0.9390)	2.1839** (1.0631)	2.3667** (0.9613)	2.4348*** (0.8739)	2.0833** (1.0291)
Subsidy	0.1885 (0.1281)						0.1735 (0.1347)
Export		0.0587 (0.1683)					−0.0592 (0.1535)
ForeignTariff			0.0969 (0.6169)				0.1962 (0.6149)
RCA				0.6295 (0.5820)			0.5842 (0.6366)
TFP					0.5887 (0.6013)		0.4181 (0.6614)
Liquidity						1.9707 (2.3009)	1.3750 (2.2915)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	580	580	580	580	580	580	580

Note: Robust standard errors clustered at ISIC four-digit industry. Controls include logAge, logN, HHI, and India&Brazil.

** p < .05.

*** p < .01.

5.2.2. Robustness

We now turn to other concerns about factors that might affect the relationship between trade liberalization and vertical integration. We address them one by one below and report the results in Table 3. We find that our main results are robust to all those considerations.

5.2.2.1. Subsidy. Chinese government has been active in promoting industrialization using industry policies such as government subsidy (Aghion et al., 2015). The coverage of firms receiving government subsidy varies across years. For example, based on the ASIFP data, Aghion et al. (2015) document that 9.4% of the firms received subsidies in 1998. The ratio increases steadily to a peak of 15.1% in 2004 and then decreases to 12.4% in 2007. Subsidy could affect firms' M&A decisions. To rule out the potential confounding effect of the subsidy policy, we control for the log of total subsidy of each industry, $Subsidy_{it}$. The results are reported in column (1) of Table 3.

5.2.2.2. Export expansion. Along with tariff cuts by China upon the WTO accession, some other countries might also liberalize their markets to Chinese exports, which implies expanded market opportunities for Chinese firms. This in turn may also affect the incentives of Chinese domestic vertical integration. We control for export expansion opportunity and its effects in two ways. First, we add a term of logarithm of industrial total exports, $Export_{it}$, to Eq. (3). Any possible change that would affect the demand for Chinese exports would ultimately be reflected in increased exports. The regression results are in column (2). Second, we include the weighted average tariffs of all foreign countries faced by Chinese exporters, $ForeignTariff_{it}$, to capture possible market expansion opportunity. The regression results are in column (3).

5.2.2.3. Comparative advantage. Industries differ in many aspects such as comparative advantages. Firms in industries of China's comparative advantage may not be affected too much by import tariff cuts but benefit more from foreign countries' tariff cuts. To determine whether such a difference matters and whether it affects our main results, we include the revealed comparative advantage index (RCA_{it}) in the model. As common in literature, we calculate the RCA index of an industry for each year as the ratio of the share of the industry's export in China's total exports over the share of the industry's export in the world's total exports. We report the regression results in column (4).

5.2.2.4. Productivity. The impacts of trade liberalization on firms' productivity have been well documented in recent trade literature. In industrial organization literature, it is found that the efficiency gain from productivity spillover is a major motivation of vertical integration.¹⁹ Do the effects of trade liberalization on vertical integration come from the change in productivity? We check this question by controlling for the industry average productivity.²⁰ We first estimate each firm's total factor productivity (TFP) using the Olley-Pakes method, and then use them to calculate the average TFP of each ISIC four-digit industry as a proxy for industry productivity. Column (5) shows that TFP is positively correlated with the number of vertical integration in each industry, but not significant.

5.2.2.5. Upstream financial constraint. Financial constraint is one of the factors that affect firms' investment decisions. Both a firm's willingness to engage in M&A as well as its bargaining power could be affected by financial constraint. We control for the financial condition of the upstream industry to tease out its effect. Following Manova and Yu (2016), we use liquidity level to capture the average financial constraint of the upstream industry. Specifically, we define the liquidity of industry

$$i \text{ in year } t \text{ as } Liquidity_{it} = \frac{\text{total current asset}_{it} - \text{total current liability}_{it}}{\text{total asset}_{it}}.^{21}$$

The results are in column (6).

In column (7), we include all the above-discussed variables in the regression. Our findings on the key variables, i.e., input and output tariffs of the target firm's industry, remain robust.

5.2.3. Other considerations

5.2.3.1. Downstream controls. So far, we mainly control for variables in the upstream industry. However, downstream factors might also affect the incentives for vertical integration and need to be considered. As each upstream industry (i) has many downstream industries (j), we construct the downstream measure of industry i , denoted as $DownX_{it}$, as the weighted average of all downstream industries' X_{jt} : $DownX_{it} = \sum_j SupplyShare_{ij} \cdot X_{jt}$, where $SupplyShare_{ij} = X_{ij}/Y_i$, and X_{ij} is the inputs from i to j , and Y_i is the output of i . Hence, the weight, $SupplyShare_{ij}$, measures how important j is as a downstream buyer to i . We consider a series of downstream factors including downstream tariffs, the average scale, competition level, and liquidity of the downstream industries in a stepwise way. Results are reported in columns (1)–(5) of Table 4, respectively. The estimates of output and input tariffs are not qualitatively affected by these downstream controls.

5.2.3.2. Alternative measures and estimation methods. We check whether our findings are robust to (a) measuring vertical integration by value, (b) defining vertical integration based on IO Table, and (c) defining vertical integration based on pair-wise upstreamness proposed by Alfaro et al. (2019). We also check the results with alternative estimation methods like Poisson Pseudo-Maximum Likelihood model, random effect Poisson model, fixed effect Negative Binomial model and Tobit model. Our main findings are robust to all those tests. We report all the results in Table A3 in the online Appendix.

5.3. Mechanism tests

Our theoretical framework emphasizes the importance of relationship-specific investment in understanding the effects of input and output tariff changes on firm organizational choices. In this subsection, we conduct further empirical analysis to determine whether the mechanism from the theoretical analysis is relevant in the reality (in our data). To this end, we first derive two hypotheses.

The value of relationship-specific investment is more important in industries where goods are more differentiated. Production of differentiated goods requires differentiated inputs. In contrast, if the inputs bought from the supplier and that imported from overseas are homogenous, the buyer will simply buy from the source with a lower price. In the latter case, the organizational decision is simple and the change in tariffs will not play an important role in determining the equilibrium outcome of the organizational form. According to Nunn (2007), if a product is sold on an organized exchange or with reference price, then its market is thick and the scope of hold-up and underinvestment is limited. Thus, we expect to have the following hypothesis:

Hypothesis 1. *The effects of tariff changes on firm's organizational choice are stronger in differentiated-goods industries than homogenous-goods industries.*

Underinvestment problem exists because the supplier and buyer cannot write a complete contract to specify the investment level and payment. When the contract is incomplete, firms rely on *ex post* bargaining. Tariffs affect the outside option of the bargaining. Thus,

¹⁹ See Alfaro et al. (2016) for a brief summary.

²⁰ Productivity is correlated with profitability. Hence, controlling for productivity also helps alleviate another concern that trade liberalization affects vertical integration through changing the profitability of firms.

²¹ Leverage level $\frac{\text{total current liability}_{it}}{\text{total current asset}_{it}}$ is another commonly used measure of financial conditions. Our results are robust to this measure as well.

Table 4
Downstream controls.

	(1) Price	(2) Scale	(3) Competition	(4) Liquidity	(5) All
WTO·OutT01	−2.0545*** (0.7011)	−1.9026*** (0.7360)	−1.9722*** (0.7382)	−1.9094*** (0.7278)	−2.0839*** (0.7130)
WTO·InT01	2.6314*** (0.8462)	2.3815*** (0.8660)	2.5168*** (0.9147)	2.4337*** (0.8836)	2.6339*** (0.8405)
Liquidity	2.2339 (2.2789)	1.7290 (2.2662)	2.1840 (2.2751)	2.0440 (2.3475)	1.6435 (2.4201)
WTO·DownT01	−0.4602 (0.2827)				−0.0003 (0.0005)
DownLogN		−0.0004 (0.0004)			7.3478 (19.9176)
DownHHI			14.5187 (18.3148)		5.0716 (5.7799)
DownLiquidity				−0.8021 (4.5576)	−0.0003 (0.0005)
Controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	580	580	580	580	580

Note: Robust standard errors clustered at ISIC four-digit industry. Controls include logAge, logN, HHI, India&Brazil and upstream liquidity.

*** $p < .01$.

underinvestment problem is more serious in an environment in which writing complete contracts is more difficult, and we have the following hypothesis:

Hypothesis 2. *The effects of tariff changes on firm's organizational choice are stronger in industries with lower contractibility.*

We conduct further empirical analyses to test the above two hypotheses.

5.3.1.1. Homogenous vs. heterogenous industries. Rauch (1999) classifies SITC four-digit products into three groups: goods traded on an organized exchange, goods traded with reference price, and differentiated goods. We group Rauch's classification into two types of industries: the homogenous industry, which mainly produces the first two types of goods in Rauch (1999), and the heterogenous industry, which produces differentiated goods. We match SITC four-digit products to ISIC four-digit industries based on the concordance provided by UNSTAT.

We run regressions based on Eq. (3), using the subsamples of heterogenous and homogenous industries separately, and report the results in columns (1) and (2) of Table 5. In heterogenous industries, the effect of output tariff is significant and negative, and the effect of input tariff is significant and positive. The effects are much stronger than the average effects obtained in Section 4 using the full sample. In homogenous industries, the effects are both significant at 10% level, but quantitatively much smaller. These results support Hypothesis 1.

Table 5
Tests of the hypotheses.

	(1) Differentiated	(2) Homogenous	(3) High volatility	(4) Low volatility
WTO·OutT01	−3.2705** (1.4980)	−1.8164* (0.9537)	−3.1804** (1.2380)	−1.8702** (0.8235)
WTO·InT01	6.5967* (3.5171)	1.7267* (0.9904)	5.2824*** (1.5450)	1.0482 (1.7229)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	270	290	290	290

Note: Robust standard errors clustered at ISIC four-digit industry. Controls include all basic and robustness controls, including logAge, logN, HHI, India&Brazil.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

5.3.1.2. Contractibility. Contractibility is affected by market uncertainty. The general argument for incomplete contract is that contracts cannot specify all possible contingencies that firms might face. We use the supplier's sales volatility to proxy uncertainty, which represents contractibility. Higher uncertainty means lower contractibility.

We measure sales volatility by average variation (ratio of standard deviation to mean) of the supplier's annual sales in the pre-WTO period, 1998–2001. We divide our sample into two subsamples: high volatility industries, which have sales volatility larger than the median, and low volatility industries, which have sales volatility lower than the median. We run regressions based on these two subsamples, and report the results in columns (3) and (4) of Table 5, respectively. The effects of tariffs on integration are evidently more pronounced in high-volatility industries than in low-volatility industries, which lends a support to Hypothesis 2.²²

6. Conclusion

International trade and industrial reorganization are inter-related. In this study, we propose a model in which vertically related firms make relationship-specific investments and trade liberalization affects their incentive for vertical integration. We empirically investigate the effects of tariff changes on the domestic organizational structure between the upstream and downstream industries. We utilize the tariff changes caused by China's accession to the WTO as an exogenous shock and use the DID method to identify the causal effects of trade liberalization on domestic vertical integration. We show that vertical integration is more likely to occur in industries where large tariff reductions on inputs of the upstream industry have occurred, whereas tariff reductions on the upstream industry's outputs reduce domestic vertical integrations. These empirical findings are consistent with the model predictions.

Existing empirical studies in the related literature have intensively examined the effects of trade liberalization on firm performances such as productivity, product quality, innovation, and entry and exit in the domestic and foreign markets. The present paper explores a different aspect: effects of trade liberalization on domestic industry's organizational structure, similar to Ornelas and Turner (2008, 2012) and Alfaro

²² Import intensities might also influence the effects of tariffs, and thus generate heterogeneous responses across industries. Indeed, we find evidence for these responses and present it Table A4 in the online appendix.

et al. (2016). A related and important issue, which has yet to be investigated, is how the resulting changes in industrial organization in general, and vertical integration in particular, affect firm productivity, industry efficiency, and social welfare. Although these issues have been studied, they are only in the context in which industrial organizational structure is held unchanged. Our findings imply that trade liberalization affects equilibrium market structure.

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Appendix A. Supplementary data

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